

# **ERP Requirement Document**

## **Job Launching & Job Movement Flow**

PCB Job Tracking Module — Job Card Launch to Packing

## Objective

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Develop a **simple and user-friendly Job Launching & Job Movement module** in the ERP system for monitoring PCB jobs from Job Card Launch to Packing.

The system should provide:

- Job Card launching
- Stage-wise job movement
- Complete visibility of all jobs and their current stages
- Full and partial/uncompleted job movement
- Barcode-based job movement
- Stage-wise user rights
- Daily job movement and WIP reporting
- Android/mobile app support

**Important: The UI should remain simple, clean and easy to operate. No unnecessary features or complicated screens should be added at this stage.**

## 1. Job Card Launching

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Create a dedicated **"New Job Card Launch"** page. On this page, there should be only one main option: **"ADD NEW JOB CARD"**.

After clicking this option, a form should open with the following fields:

| Field                           | Example / Requirement          |
|---------------------------------|--------------------------------|
| Job Card Photo                  | Attach Job Card Photo          |
| Job Card No.                    | 26-27-1729                     |
| Customer Part No.               | EV-900W-WP-TO247-VORS-25082026 |
| R.F.E. Part Code                | D3625                          |
| Customer Code                   | CUST-RF045                     |
| Target Date                     | 08-Sep-26                      |
| Priority                        | MOST URGENT                    |
| Total Production PNL Qty        | 40                             |
| Total Customer PNL Qty          | 80                             |
| Total PCB Qty                   | 160                            |
| Total Production PNL Area (Sqm) | 50                             |
| Total Customer PNL Area (Sqm)   | 45                             |
| Job Flow Selection              | PF-01                          |

## Job Flow Selection

"Job Flow Selection" should be a **dropdown menu**. Example: **PF-01**. When PF-01 is selected, the system should follow the predefined process flow below:

| Process No. | Process / Stage |
|-------------|-----------------|
| 1           | SHEARING        |
| 2           | DRILLING        |
| 3           | DRL-QC          |
| 4           | DML             |
| 5           | PIT             |
| 6           | PIT-QC          |
| 7           | PLATING         |
| 8           | ETCHING         |
| 9           | PREMASK-QC/AOI  |
| 10          | PISM            |
| 11          | PISM-QC         |
| 12          | HASL            |
| 13          | HASL-QC         |
| 14          | LEGEND PRINT    |
| 15          | ROUTING & VG    |
| 16          | BBT             |
| 17          | FQC (AI)        |
| 18          | PDI-AQL         |
| 19          | PACKING         |

After entering all required information, there should be a **"LAUNCH JOB"** option. Once the job is launched, it should automatically appear in the **first process/stage** of the selected Job Flow.

## 2. Job Update / Job Movement Page

Create a **Job Update** page where all active jobs can be viewed. The purpose of this page is to clearly show:

- Which jobs are currently running
- Which stage each job is currently at
- Job number
- Part code
- Customer
- Quantity
- Area (Sqm)
- Target date
- Priority
- Current process/stage
- Delay/overdue status

The layout/design should be similar to the reference photo provided separately.

The UI should be:

- Simple
- Clean
- Easy to understand
- Easy to operate
- Suitable for desktop ERP
- Also suitable for Android/mobile application

### 3. Job Selection and Movement Options

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When a user clicks on a particular job, for example **Job Card No.: 26-27-1729**, the system should show three options:

#### A. Job Card View

- The Job Card Photo uploaded during Job Launch should be displayed.
- The user should be able to verify the original Job Card.

#### B. Full Job Movement

When the user selects Full Job Movement, a confirmation window should appear. Example:

*"Are you sure you want to move Job Card No. 26-27-1729 to the next process?"*

The user should be able to confirm the movement. There should also be a **Remark** field where the user can enter remarks related to:

- Rejection
- Rework
- Process issue
- Other relevant movement remarks

The job should then move completely to the next process. Example: If Job Card No. 26-27-1729 has 40 Production PNL and Full Job Movement is selected, the complete 40 PNL should move to the next stage.

### 4. Uncompleted / Partial Job Movement

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There should be a separate option: **C. Uncompleted Job Movement**. This option is required when the complete job is not ready to move to the next process.

Example: Total job quantity = 40 PNL. If only 35 PNL is ready to move forward:

- 35 PNL should move to the next process.
- The complete available data/details for those 35 PNL should be visible at the next process.
- The remaining 5 PNL should stay at the current/previous process.
- The system should maintain the balance quantity automatically.

Example Flow — Current Stage: DRILLING, Total = 40 PNL, Move = 35 PNL. After movement:

| Stage    | Result          |
|----------|-----------------|
| DRILLING | 5 PNL remaining |

| Stage  | Result       |
|--------|--------------|
| DRL-QC | 35 PNL moved |

The system should maintain the correct quantity and area information for both portions.

## 5. Barcode-Based Job Movement

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In addition to manually selecting/entering the Job Card Number, the system should provide a **barcode scanning option**. Each Job Card will have a barcode containing the Job Card Number. Example: Barcode contains 26-27-1729.

When the user scans the barcode:

- The system should identify the Job Card.
- The corresponding job should open immediately.
- The available Job Movement options should be displayed.
- The user should be able to select: Job Card View / Full Job Movement / Uncompleted Job Movement.

The objective is to make job movement faster and reduce manual Job Card Number entry.

## 6. Stage-Wise User Rights / Access Control

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Each production stage should have its own user rights.

**Important Requirement: A user assigned to one stage should NOT be able to move jobs from another stage.**

For example:

- SHEARING user → Can move SHEARING jobs only
- DRILLING user → Can move DRILLING jobs only
- PLATING user → Can move PLATING jobs only
- HASL user → Can move HASL jobs only
- PACKING user → Can move PACKING jobs only

Users should not be able to perform job movement from stages for which they do not have permission. The system should therefore have stage-wise access rights.

## 7. Daily Job Movement & WIP Report

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The system should generate a daily report based on all job movements made during the day. The report should provide a clear summary of production status.

### Stage-Wise Job Status

- Total number of jobs at each stage
- Total PNL quantity at each stage
- Total Sqm at each stage
- Current WIP Sqm at each stage

### Delay Monitoring

- Overdue jobs
- Over-delayed jobs
- Jobs exceeding their Target Date

### Quality / Loss Monitoring

- Total Rework during the day
- Total Rejection during the day

*Objective: quickly understand how many jobs are at each stage, how much WIP is available, which jobs are delayed, and how much rework/rejection occurred during the day.*

## 8. Basic Process Flow

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- 1 Job Card Launch
- 2 Select Job Flow (PF-01)
- 3 Job automatically appears at first process
- 4 Stage User opens Job
- 5 Job Card View / Full Movement / Uncompleted Movement
- 6 Job moves to next stage
- 7 Next Stage User receives the job
- 8 Process continues until PACKING
- 9 Daily Job Movement & WIP Report

## 9. Android Application

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The same Job Launching and Job Movement functionality should also be available through an **Android application**. The Android app should support the essential functions required for shop-floor operation, especially:

- Viewing assigned jobs
- Viewing current job status
- Job Card View
- Full Job Movement

- Uncompleted Job Movement
- Barcode scanning
- Remarks entry
- Stage-wise access control

The mobile interface should remain simple and easy to use on the production floor.

## 10. Important Development Instructions

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- 1 Keep the UI simple and clean.
- 2 Do not add unnecessary features at this stage.
- 3 The main focus should be Job Launching + Job Movement + Job Status + Daily Report.
- 4 Every job must have a clear current-stage status.
- 5 Full and partial/uncompleted movement must be supported.
- 6 Quantity and Sqm data must move correctly with the job.
- 7 Remaining quantity must automatically stay at the previous stage in case of partial movement.
- 8 Barcode scanning must be available for quick job identification and movement.
- 9 Stage-wise user rights must be strictly implemented.
- 10 A user must not be able to move a job from another stage without authorization.
- 11 The system should maintain proper movement history/data for reporting.
- 12 The same basic functionality should be available on Android.
- 13 The first version should be developed only according to the above requirements.
- 14 Do not add extra modules or complicated functionality until this basic flow is tested and approved.

## Initial Development Scope

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For the first development/testing phase, please complete only:

- 1 Job Card Launch
- 2 Job Update / Job Status Page
- 3 Full Job Movement
- 4 Uncompleted / Partial Job Movement
- 5 Job Card Photo View
- 6 Barcode-Based Job Movement
- 7 Stage-Wise User Rights
- 8 Daily Job Movement & WIP Report
- 9 Android App Basic Functionality

After completing the above, we will test the complete flow in actual production conditions. Once the basic system is working correctly, we will discuss and add the remaining updates/features.

# User ID & Access Rights

The ERP system should have **three levels of User IDs**, with clearly defined permissions.

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## 1. Master ID

The Master ID will have complete system access and full administrative rights.

### *Permissions:*

- Access to all activities and all modules
- Add / Launch new Job Cards
- View all jobs at every stage
- Move jobs from any stage
- Full Job Movement
- Uncompleted / Partial Job Movement
- View Job Card Photo
- Barcode-based Job Movement
- Delete Job Card from any stage
- Access to all reports
- Access to all WIP and movement data
- Create / modify / manage user IDs
- Assign stage-wise rights to users
- Modify stage/process configuration
- Modify Job Flow configuration
- Manage all system configurations

**Master ID = Complete ERP Control**

## 2. Super User

The Super User will have operational access across the complete production flow, but will not have Master-level configuration/deletion rights.

### *Permissions:*

- Add / Launch new Job Cards
- View all jobs
- View jobs at all stages
- Move jobs from any production stage
- Full Job Movement
- Uncompleted / Partial Job Movement
- Job Card View
- Barcode-based Job Movement
- Enter movement/rework/rejection remarks
- View all job movement status
- View WIP and daily reports

### *Restriction:*

- Cannot delete jobs
- Cannot change Master/System configuration
- Cannot create or modify Master-level system settings

**Super User = Job Launch + Complete Production Movement Control**

### 3. Normal User

The Normal User will have access only to the production stage(s) specifically assigned to that User ID.

#### *Permissions:*

- View jobs assigned to the selected stage
- Move jobs from the assigned stage only
- Full Job Movement
- Uncompleted / Partial Job Movement
- Job Card View
- Barcode-based Job Movement
- Enter movement/rework/rejection remarks for their assigned stage

#### *Restrictions:*

- Cannot move jobs from any other stage
- Cannot add/launch new Job Cards
- Cannot delete Job Cards
- Cannot change Job Flow
- Cannot change user rights
- Cannot access system configuration

Example: If a Normal User is assigned to PLATING, then that user can view PLATING jobs and move PLATING → Next Stage. But the same user cannot move DRILLING → DRL-QC or HASL → HASL-QC or any other stage unless that stage is specifically assigned to the User ID.

### User Rights Summary

| Activity                       | Master ID | Super User | Normal User |
|--------------------------------|-----------|------------|-------------|
| Add / Launch Job Card          | Yes       | Yes        | No          |
| View All Jobs                  | Yes       | Yes        | No          |
| Move Job – Any Stage           | Yes       | Yes        | No          |
| Move Job – Assigned Stage      | Yes       | Yes        | Yes         |
| Full Job Movement              | Yes       | Yes        | Yes         |
| Uncompleted / Partial Movement | Yes       | Yes        | Yes         |
| Job Card View                  | Yes       | Yes        | Yes         |
| Barcode Job Movement           | Yes       | Yes        | Yes         |
| Enter Rework/Rejection Remarks | Yes       | Yes        | Yes         |
| Delete Job Card                | Yes       | No         | No          |

| Activity                     | Master ID | Super User | Normal User  |
|------------------------------|-----------|------------|--------------|
| View Reports                 | Yes       | Yes        | As permitted |
| User Management              | Yes       | No         | No           |
| Stage/Job Flow Configuration | Yes       | No         | No           |
| System Configuration         | Yes       | No         | No           |

## Important Security Requirement

The system must enforce these rights at the backend/database level, not only by hiding buttons from the UI. For example, even if a Normal User tries to access another stage through a direct URL, API call, or any other method, the system must block the action if that stage is not assigned to that User ID.